

PETRA: A first-in-class, first-in-human trial of the next-generation PARP1-selective inhibitor AZD5305 in patients with *BRCA1/2*, *PALB2* or *RAD51C/D* mutations

CT007

Speaker

Timothy A. Yap, MD

University of Texas MD Anderson Cancer Center, TX, USA

Authors

Timothy A. Yap¹, Seock-Ah Im², Alison M. Schram³, Adam Sharp⁴, Judith Balmana⁵, Richard D. Baird⁶, Jessica S. Brown⁷, Maria Schwaederle⁸, Elizabeth A. Pilling⁹, Ganesh Moorthy¹⁰, Spiros Linardopoulos¹¹, Adam Dowson¹², Carol Pound¹³, Edit Lukacs¹⁴, Sabina Cosulich¹⁵, Stephen J. Luen¹⁶

¹University of Texas MD Anderson Cancer Center, TX, USA; ²Seoul National University Hospital, Cancer Research Institute, Seoul National University College of Medicine, Seoul, Republic of Korea; ³Memorial Sloan Kettering Cancer Center, New York, NY, USA; ⁴The Institute of Cancer Research, The Royal Marsden, London, UK; ⁵Vall d'Hebron University Hospital and Vall d'Hebron Institute of Oncology (VHIO), Barcelona, Spain; ⁶Cancer Research UK Cambridge Centre, Cambridge, UK; ⁷Early Global Development, Oncology R&D, AstraZeneca, Cambridge, UK; ⁸Early Oncology Clinical Development, Oncology R&D, AstraZeneca, Gaithersburg, MD, USA; ⁹Oncology Biometrics, Oncology R&D, AstraZeneca, Cambridge, UK; ¹⁰Clinical Pharmacology and Quantitative Pharmacology, R&D, AstraZeneca, Boston, MA, USA; ¹¹Translational Medicine, Oncology R&D, AstraZeneca, Cambridge, UK; ¹²Early Global Development, Oncology R&D, AstraZeneca, Cambridge, UK; ¹³Early Global Development, Oncology R&D, AstraZeneca, Gaithersburg, MD, USA; ¹⁴Global Patient Safety, AstraZeneca, Cambridge, UK; ¹⁵Oncology R&D, AstraZeneca, Cambridge, UK; ¹⁶Peter MacCallum Cancer Centre, University of Melbourne, Melbourne, Australia

Disclosures



APRIL 8-13 • #AACR22

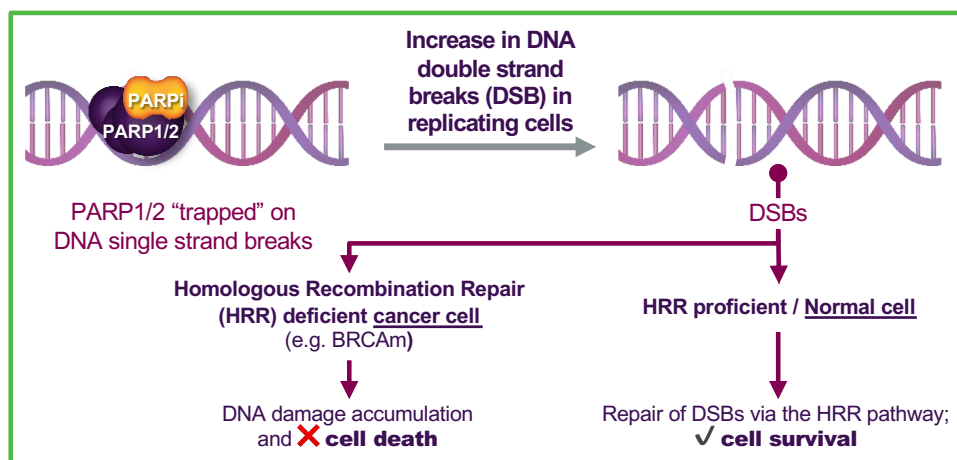
Dr. Yap reports:

- **Employment:** University of Texas MD Anderson Cancer Center, where I am Medical Director of the Institute for Applied Cancer Science, which has a commercial interest in DDR and other inhibitors (e.g. IACS30380/ART0380 was licensed to Artios).
- **Grant/Research support (to the Institution):** Acrivon, Artios, AstraZeneca, Bayer, Beigene, BioNTech, Blueprint, BMS, Clovis, Constellation, Cyteir, Eli Lilly, EMD Serono, Forbuis, F-Star, Artios, GlaxoSmithKline, Genentech, Haihe, ImmuneSensor, Ionis, Ipsen, Jounce, Karyopharm, KSQ, Kyowa, Merck, Mirati, Novartis, Pfizer, Ribon Therapeutics, Regeneron, Repare, Rubius, Sanofi, Scholar Rock, Seattle Genetics, Tesaro, Vivace and Zenith.
- **Consultant for:** AbbVie, AstraZeneca, Acrivon, Adagene, Almac, Aduro, Amphista, Artios, Athena, Atrin, Avoro, Axiom, Baptist Health Systems, Bayer, Beigene, Boxer, Bristol Myers Squibb, C4 Therapeutics, Calithera, Cancer Research UK, Clovis, Cybrexa, Diffusion, EMD Serono, F-Star, Genmab, Glenmark, GLG, Globe Life Sciences, GSK, Guidepoint, Idience, Ignyta, I-Mab, ImmuneSensor, Institut Gustave Roussy, Intellisphere, Jansen, Kyn, MEI pharma, Mereo, Merck, Natera, Nexys, Novocure, OHSU, OncoSec, Ono Pharma, Pegascy, PER, Pfizer, Piper-Sandler, Prolynx, Repare, resTORbio, Roche, Schrodinger, Theragnostics, Varian, Versant, Vibliome, Xinthera, Zai Labs and ZielBio.
- **Stockholder in:** Seagen

AZD5305 is a PARP1 selective inhibitor and trapper

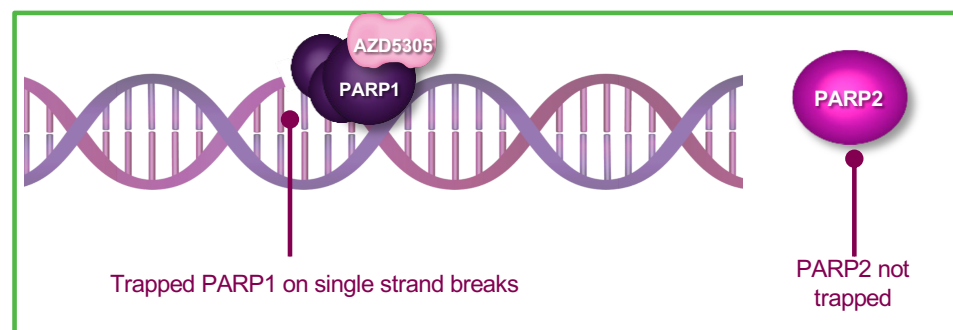
Mechanism of action of PARP inhibitors¹

Efficacious monotherapy PARP inhibitors trap PARP1 and PARP2 on DNA single strand breaks and selectively kill cancer cells



- First generation PARP inhibitors are approved in multiple indications and are dual PARP1-PARP2 inhibitors/DNA-trappers
- The main clinical adverse events observed with 1st generation PARPi are hematological toxicities
- Pre-clinically, PARP2 is essential for erythropoiesis²

AZD5305 is selective for PARP1

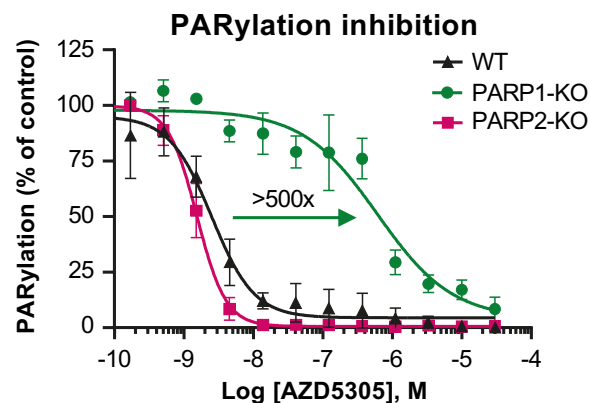


- Only PARP1 inhibition/ trapping is required for synthetic lethality in HRR deficient settings^{3,4}
- A selective PARP1 inhibitor is expected to:
 - Have a favorable tolerability profile
 - Enable higher drug exposure over target
 - Achieve greater and more durable target inhibition
 - Achieve greater antitumor efficacy
 - Enable broader combination options

AZD5305 was discovered as a result of innovative structure- and property-based design¹ to be more potent and efficacious than first generation PARPi

Highly selective for PARP1 vs PARP2

- AZD5305 is highly selective (>500-fold) for PARP1 vs PARP2²

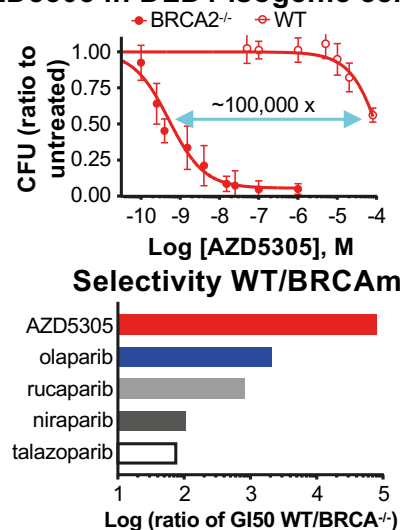


- AZD5305 is highly selective against a panel of secondary pharmacology targets

Highly selective for homologous recombination deficiency

- HRR deficient cells are 100,000-fold more sensitive to AZD5305 than normal cells³

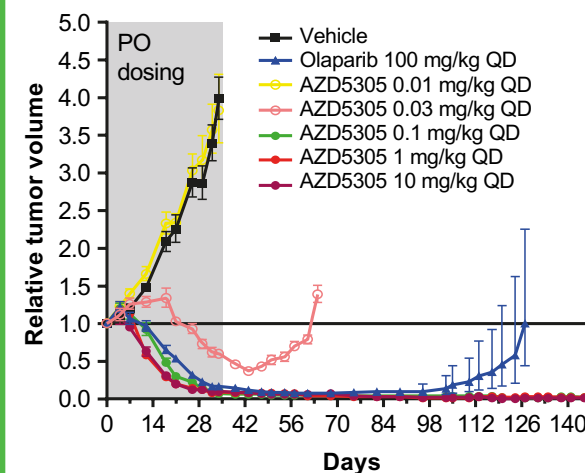
AZD5305 in DLD1 isogenic cell line pair



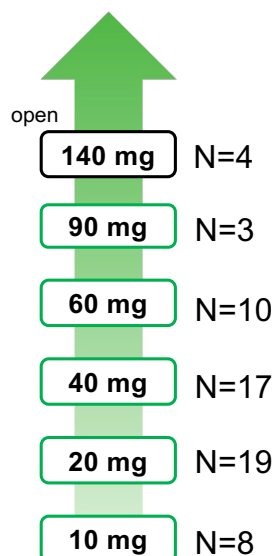
Sufficient to drive deep and durable responses

- Durable regression observed following cessation of AZD5305 dosing⁴

MDA-MB-436 (BRCA1m)



First in human PETRA study



Global Phase 1/2 Modular study

NCT04644068
(Recruiting)

Key inclusion criteria:

- Pts ≥ 18 years old with progressive advanced/metastatic ovarian, *HER2*^{neg} breast, pancreatic, or prostate cancer
- Loss of function mutation in ***BRCA1*, *BRCA2*, *PALB2*, *RAD51C* or *RAD51D*** (blood or tissue local testing; retrospective tissue central confirmation)
- **Up to one prior line of PARPi**
- Platinum-sensitive or resistant/refractory permitted
- **ECOG PS 0–2**
- **Hgb ≥ 9 g/dL**, platelets $\geq 100 \times 10^9/L$, ANC $\geq 1.5 \times 10^9/L$

Primary endpoints

- Safety and tolerability

Other endpoints

- Pharmacokinetics (PK)
- Pharmacodynamics (PD) in PBMCs and paired tumor samples
- Preliminary efficacy
- ctDNA analysis (baseline, longitudinal and at progression)

- First presentation of clinical data for this ongoing study:
 - First patient dosed Nov 25, 2020
 - **Data cutoff date: Feb 22, 2022**
 - **Monotherapy dose escalation: 61 patients dosed from 10 mg to 140 mg QD**
 - Part A dose-escalation cohorts were expanded for further evaluation of PK, PD, safety, tolerability and biological activity, once the dose had been declared as tolerated by the SRC

Demographics and disease characteristics



APRIL 8-13 • #AACR22

Characteristics	Total (N=61)
Median age (range), yrs	56.0 (25, 78)
Age ≥65 years, n (%)	16 (26.2)
Male / female, n (%)	20 (32.8) / 41 (67.2)
Race, n (%)	
White	40 (65.6)
Asian	11 (18.0)
Black/African American	3 (4.9)
Other	4 (6.6)
Pending ¹	3 (4.9)
ECOG PS 0 / 1, n (%)	25 (41.0) / 36 (59.0)
Tumor type, n (%)	
Ovarian	19 (31.1)
Platinum sensitive (PD ≥ 6 mos post-platinum)	4 (21.1)
Platinum Resistant/Refractory	14 (73.7)
Pending ¹	1 (5.3)
Breast	18 (29.5)
TNBC	5 (27.8)
HR+	12 (66.7)
Pending ¹	1 (5.6)
Pancreas	12 (19.7)
Prostate	10 (16.4)
Pending ¹	2 (3.3)

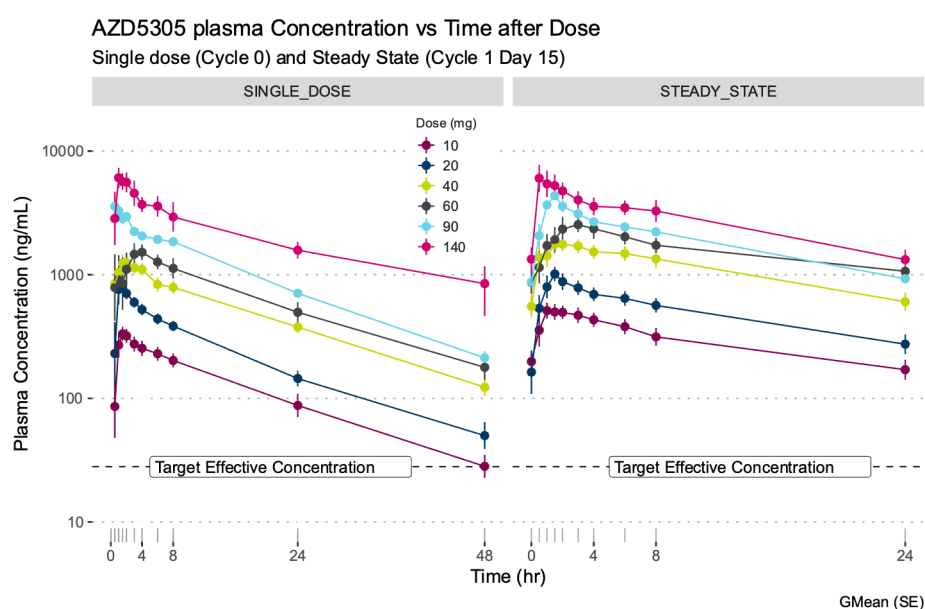
Characteristics	Total (N=61)
Alteration for eligibility (by eCRF), n (%)	
<i>BRCA1</i>	21 (34.4)
<i>BRCA2</i>	29 (47.5)
<i>PALB2</i>	7 (11.5)
<i>RAD51C</i>	1 (1.6)
<i>RAD51D</i>	0
Pending ¹	3 (4.9)
Median number of prior lines of therapy (range)	3 (0, 8)
Prior PARPi / platinum therapy status, n (%)	
PARPi and/or platinum	44 (72.1)
Both PARPi and platinum	18 (40.9)
Platinum only	20 (45.5)
PARPi only	6 (13.6)
No PARPi or platinum	17 (27.9)

¹Data entry pending at DCO

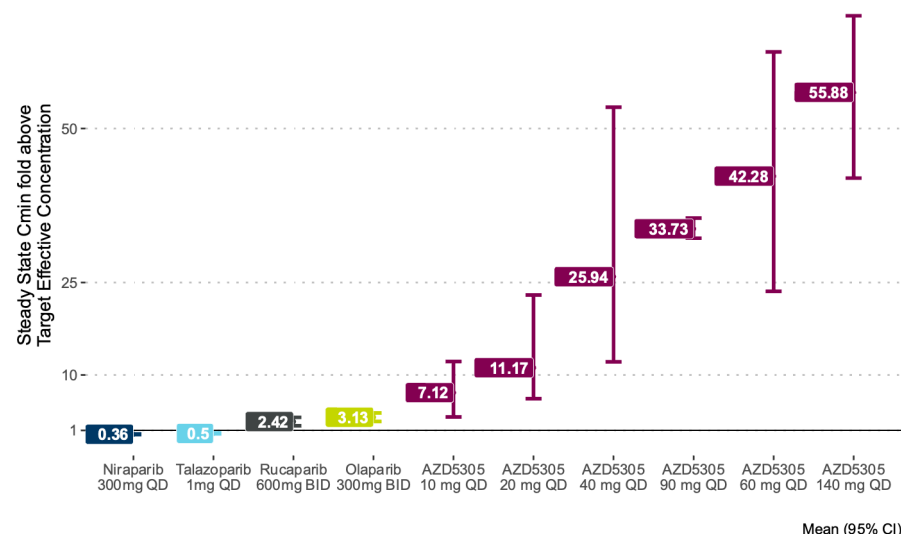
eCRF, electronic case report form; HR, hormone receptor; TNBC, triple-negative breast cancer

AZD5305 achieves higher fold coverage over the target effective concentration compared to first generation PARPi

AZD5305 plasma concentration vs time after dose



First-generation PARPi and AZD5305 C_{min} fold above TEC



- Dose proportional increase in exposure (C_{max} & AUC) observed with increase in dose (10 – 140 mg QD)
- Quick onset (T_{max} 0.5-3 h) with mean terminal elimination half-life 13.1–16.4 h across cohorts
- Steady state C_{min} above target effective concentration (TEC) in all patients with mean fold C_{min}/TEC 7.12 and 55.88 at 10 mg and 140 mg QD, respectively

Results: Overall safety summary



APRIL 8-13 • #AACR22

AE category, n (%)	AZD5305 10 mg QD (n=8)	AZD5305 20 mg QD (n=19)	AZD5305 40 mg QD (n=17)	AZD5305 60 mg QD (n=10)	AZD5305 90 mg QD (n=3)	AZD5305 140 mg QD (n=4)	Total (N=61)
Duration of therapy (months), median (range)	4.8 (1.8-14.9)	2.8 (0.9-11.5)	2.0 (0.4-9.6)	1.95 (0-3.8)	4.8 (4.6-4.9)	0.7 (0.6-1.6)	2.1 (0-14.9)
All TEAEs	8 (100.0)	19 (100.0)	17 (100.0)	8 (80.0)	3 (100.0)	0	55 (90.2)
Grade ≥3 AEs	3 (37.5)	10 (52.6)	6 (35.3)	3 (30.0)	2 (66.7)	0	24 (39.3)
Serious TEAEs	0	9 (47.4)	5 (29.4)	2 (20.0)	0	0	16 (26.2)
Discontinuations	0	0	0	1 (10.0)	0	0	1 (1.6)
Dose reductions	1 (12.5)	0	1 (5.9)	0	0	0	2 (3.3)
AZD5305-related TEAEs	5 (62.5)	11 (57.9)	9 (52.9)	4 (40.0)	3 (100.0)	0	32 (52.5)
Grade ≥3 AEs	3 (37.5)	3 (15.8)	1 (5.9)	0	2 (66.7)	0	9 (14.8)
Serious TEAEs	0	0	0	0	0	0	0
Discontinuations	0	0	0	0	0	0	0
Dose reductions	1 (12.5)	0	1 (5.9)	0	0	0	2 (3.3)

- AZD5305 was well tolerated across doses
- Only 2 patients (3.3%) had dose reductions (for grade 3 neutropenia and grade 1 thrombocytopenia related to AZD5305)
- At DCO there have been no DLTs and no AZD5305-related serious AEs or treatment discontinuations

TEAEs occurring in $\geq 10\%$ of patients overall



APRIL 8-13 • #AACR22

TEAE preferred term, (%) [*]	AZD5305 10 mg QD (n=8)		AZD5305 20 mg QD (n=19)		AZD5305 40 mg QD (n=17)		AZD5305 60 mg QD (n=10)		AZD5305 90 mg QD (n=3)		AZD5305 140 mg QD (n=4)		Total (N=61)	
Grade	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All
Nausea	0	50.0	0	42.1	0	11.8	0	50.0	0	66.7	0	0	0	34.4
Anemia [†]	25.0	25.0	21.1	31.6	11.8	17.6	0	10.0	33.3	33.3	0	0	14.8	21.3
Constipation	0	25.0	0	36.8	0	11.8	0	10.0	0	0	0	0	0	19.7
Neutropenia [†]	12.5	37.5	5.3	15.8	0	11.8	0	10.0	66.7	66.7	0	0	6.6	18.0
Thrombocytopenia [†]	12.5	12.5	5.3	21.1	0	29.4	0	10.0	0	0	0	0	3.3	18.0
Fatigue and asthenia [†]	0	25.0	0	10.5	0	23.5	0	20.0	0	0	0	0	0	16.4
Dyspnea	0	25.0	0	10.5	0	17.6	0	0	0	0	0	0	0	11.5

- AZD5305 was well tolerated across doses
- Rates of nausea and hematological toxicity were low

Favorable safety profile of AZD5305 at higher target coverage compared to first generation PARPi across all key parameters



APRIL 8-13 • #AACR22

	Niraparib ¹		Talazoparib ²		Rucaparib ³		Olaparib ⁴		AZD5305	
	N=289		N=286		N=115		N=205		N=61	
Selective for PARP1 vs. PARP2	No		No		No		No		Yes	
Fold coverage over target effective concentration	0.36		0.5		2.42		3.13		7.12 – 55.88	
Gastrointestinal toxicity (%)										
Nausea (any grade)	58		50		52		58		34	
Vomiting (any grade)	38		27		22		32		10	
Hematological toxicity (%) by Grade (≥ 3 and All grade)	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All
Anemia*	33	54	40	55	25	44	16	40	15	21
Neutropenia*	10	19	22	36	7	10	9	27	7	18
Thrombocytopenia*	16	34	15	28	10	25	NR	NR	3	18
Tolerability (%)										
Dose reduction due to any TEAE	44		53		41		25		3	
Discontinuation due to any TEAE	14		8		8		5		2	

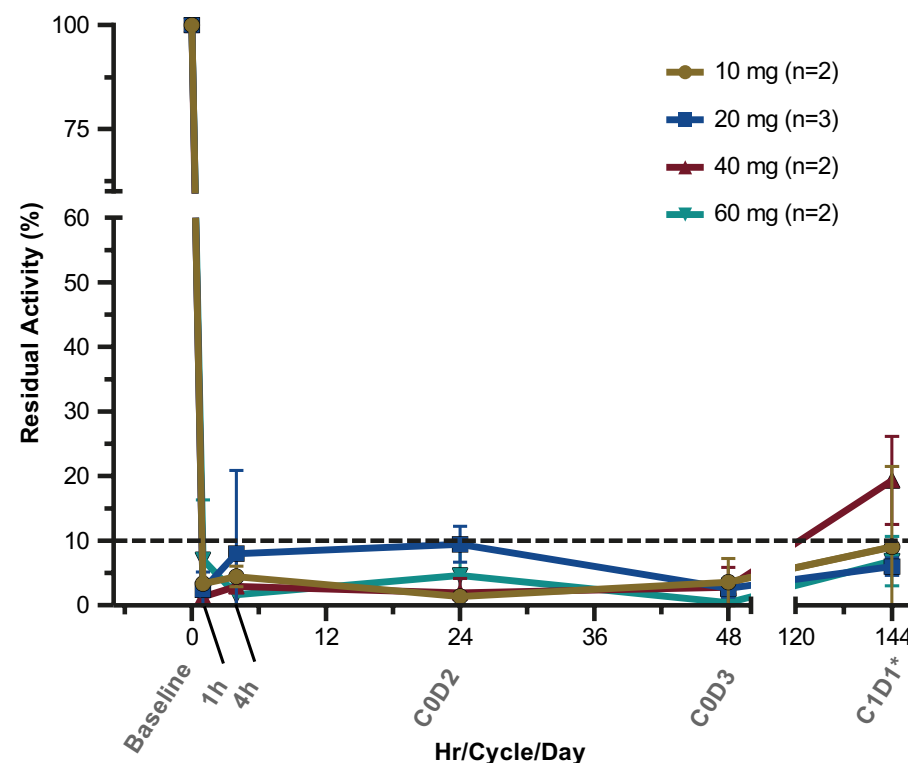
¹GALAHAD; [https://doi.org/10.1016/S1470-2045\(21\)00757-9](https://doi.org/10.1016/S1470-2045(21)00757-9) ²EMBRACA; <https://doi.org/10.1016/j.annonc.2020.08.2098> ³TRITON2 DOI: 10.1200/JCO.20.01035

⁴OLYMPIAD; DOI: <https://doi.org/10.1093/annonc/mdz012>

*Grouped terms; All AEs presented irrespective of causality; AZD5305 safety analysis set 10-140 mg; NR, not reported; TEAE, treatment emergent adverse event

PARylation inhibition data confirms robust target engagement

- PARylation inhibition has been a useful measure of target engagement for first generation PARPi
- Pharmacodynamic analyses were carried out on PBMCs to measure PARP1 inhibition by the loss of signal from poly(ADP-ribose) after a single dose of AZD5305
- Durable PARylation inhibition $\geq 90\%$ was observed at all tested doses (10, 20, 40, and 60 mg) in PBMCs of 9 patients after a single dose of AZD5305



* C1D1 was median 1 day after C0D3 (max 4 days post C0D3)

AACR **ANNUAL MEETING**
American Association for Cancer Research*
2022 *New Orleans*

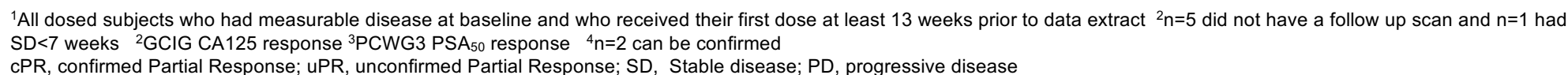
(N=40[‡])

12

AACR ANNUAL MEETING
American Association for Cancer Research*
2022 *New Orleans*

Best % change in target lesions size by tumor type

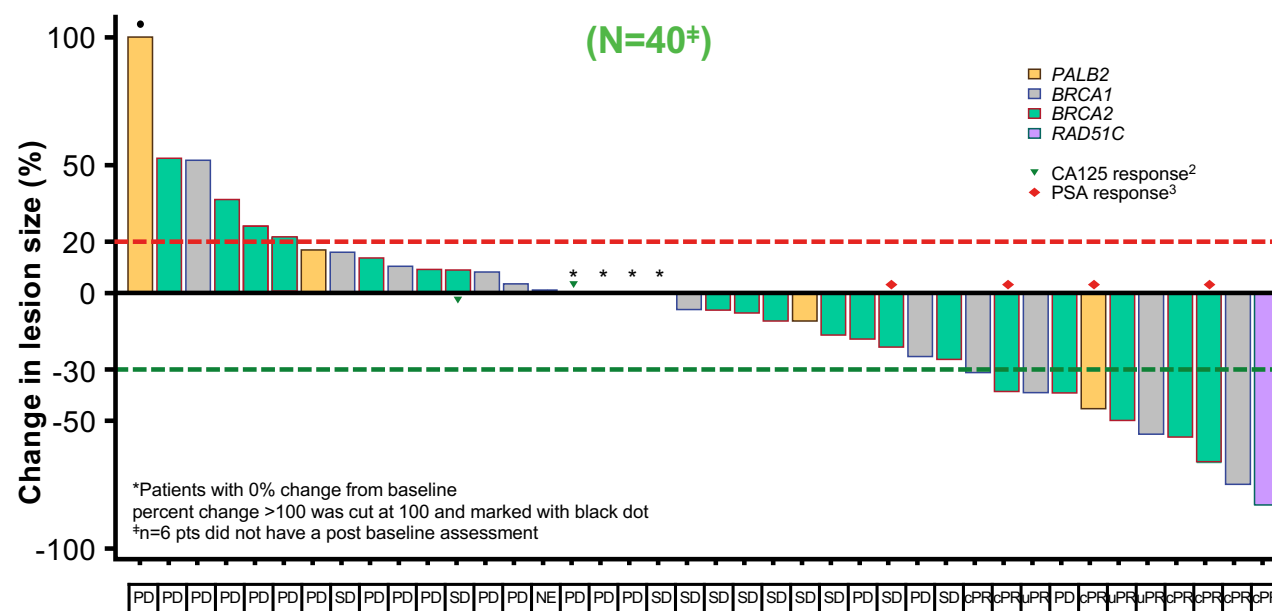
(N=40[‡])



RECIST v1.1 responses were observed starting at the lowest dose level (10 mg)

- 46 patients were included in the interim response analysis set¹
- 6 patients were not evaluable² for RECIST v1.1 assessment
- Of 40 evaluable patients, by RECIST v1.1, we observed:
10 PRs (7 confirmed; 3 unconfirmed⁴),
11 SD, and 19 PD
- Responses seen across:
 - Doses
 - Tumor types
 - Mutation types

Best % change in target lesions size by mutation

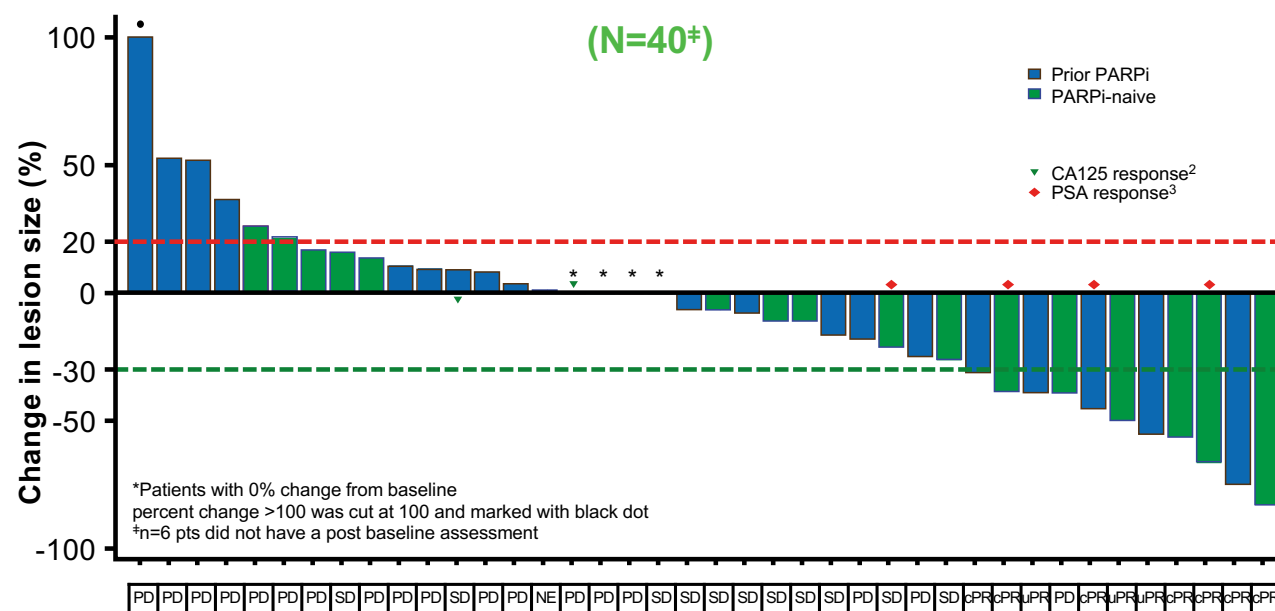


¹All dosed subjects who had measurable disease at baseline and who received their first dose at least 13 weeks prior to data extract ²n=5 did not have a follow up scan and n=1 had SD<7 weeks ³GCIG CA125 response ⁴PCWG3 PSA₅₀ response ⁴n=2 can be confirmed
cPR, confirmed Partial Response; uPR, unconfirmed Partial Response; SD, Stable disease; PD, progressive disease
Molecular status per eCRF (local testing) / 100% concordance with the retrospective central testing (MYRIAD) on the n=24 pts with results

RECIST v1.1 responses were observed starting at the lowest dose level (10 mg)

- 46 patients were included in the interim response analysis set¹
- 6 patients were not evaluable² for RECIST v1.1 assessment
- Of 40 evaluable patients, by RECIST v1.1, we observed:
10 PRs (7 confirmed; 3 unconfirmed⁴),
11 SD, and 19 PD
- Responses seen across:
 - Doses
 - Tumor types
 - Mutation types
 - and independent of prior PARPi use

Best % change in target lesions size by prior PARPi



¹All dosed subjects who had measurable disease at baseline and who received their first dose at least 13 weeks prior to data extract ²n=5 did not have a follow up scan and n=1 had SD<7 weeks ³GCIG CA125 response ⁴PCWG3 PSA₅₀ response ⁵n=2 can be confirmed
cPR, confirmed Partial Response; uPR, unconfirmed Partial Response; SD, Stable disease; PD, progressive disease

AACR ANNUAL MEETING
American Association for Cancer Research*
2022 *New Orleans*

Tumor type and dose (mg)

Weeks

Legend:

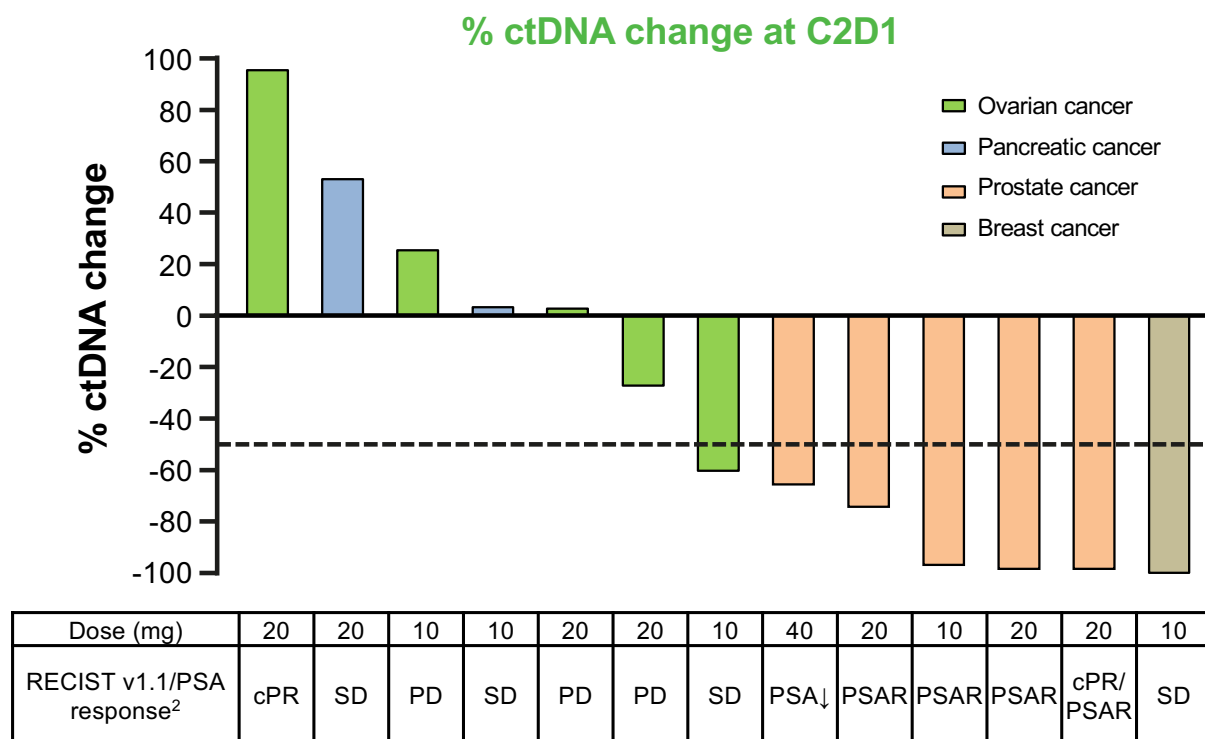
- Prior PARPi (Blue)
- PARPi-naïve (Green)
- Ongoing treatment (Orange arrow)

The chart displays the duration of treatment for various tumor types and doses. The x-axis represents weeks from 0 to 66. The y-axis lists tumor types and doses. A vertical dashed line marks week 16. The legend indicates that blue bars represent Prior PARPi, green bars represent PARPi-naïve, and orange arrows represent Ongoing treatment.

Tumor type and dose (mg)	Weeks	Treatment Type
OC 20	50	Prior PARPi
OC 20	33	Prior PARPi
OC 20	30	Prior PARPi
OC 10	28	Prior PARPi
PC 90	22	Prior PARPi
OC 60	15	Prior PARPi
OC 20	14	Prior PARPi
PaC 20	13	Prior PARPi
BC 60	12	Prior PARPi
OC 20	12	Prior PARPi
OC 40	9	Prior PARPi
OC 40	9	Prior PARPi
PaC 40	9	Prior PARPi
PaC 20	9	Prior PARPi
OC 20	8	Prior PARPi
OC 10	8	Prior PARPi
PaC 10	8	Prior PARPi
BC 40	7	Prior PARPi
OC 20	7	Prior PARPi
OC 20	6	Prior PARPi
OC 20	5	Prior PARPi
OC 20	4	Prior PARPi
BC1 40	3	Prior PARPi
OC 10	66	PARPi-naïve
BC 40	42	PARPi-naïve
PC 20	27	PARPi-naïve
OC 10	25	PARPi-naïve
BC 10	24	PARPi-naïve
PC 20	24	PARPi-naïve
PaC 40	23	PARPi-naïve
BC 90	22	PARPi-naïve
PC 90	21	PARPi-naïve
PC 40	19	PARPi-naïve
PC 60	18	PARPi-naïve
OC 10	17	PARPi-naïve
PC 40	16	PARPi-naïve
BC 40	15	PARPi-naïve
PaC 60	14	PARPi-naïve
BC 20	13	PARPi-naïve
BC 10	12	PARPi-naïve
PaC 60	11	PARPi-naïve
BC 20	10	PARPi-naïve
BC 40	9	PARPi-naïve
BC 40	8	PARPi-naïve
BC 40	7	PARPi-naïve
BC1 40	6	PARPi-naïve
PaC 60	5	PARPi-naïve
PaC 40	4	PARPi-naïve
PaC 40	3	PARPi-naïve
BC1 40	2	PARPi-naïve
60	1	PARPi-naïve
PaC 40	1	PARPi-naïve
PC 60	1	PARPi-naïve
BC 40	1	PARPi-naïve
OC 60	1	PARPi-naïve

ctDNA responses by AZD5305 treatment across different tumor types and doses

- 8 out of 13 patients showed a ctDNA decrease¹ including 5/5 prostate cancer patients



¹Mean variant allele frequency (mVAF) of all somatic variants detected with at least 0.3% allele frequency; % change was calculated from baseline to C2D1 (Guardant Health OMNI at screening and GH360 thereafter) ²In patients with no measurable disease the PCWG3 PSA₅₀ response (confirmed and unconfirmed) only is represented.

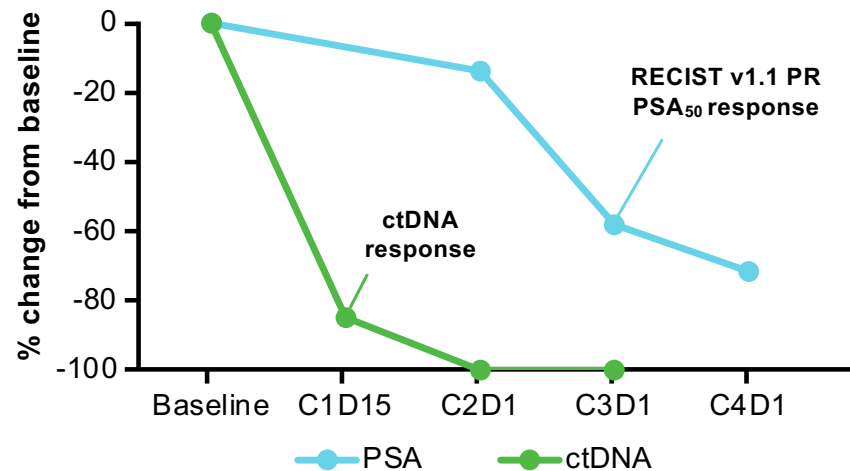
cPR, confirmed Partial Response; SD, stable disease; PD, progressive disease; PSAR, PCWG3 PSA₅₀ response; PSA↓, PSA decrease 0-49%

Early ctDNA response in a prostate cancer patient with a RECIST v1.1 confirmed partial response

ctDNA response ~45 days prior to 1st scan

Case #1: 66-year-old man with a *BRCA2m* prostate cancer (AZD5305 20 mg)

- 3 prior lines of therapy (ADT, docetaxel and abiraterone)
- RECIST v1.1 PR (-38.6%), confirmed at second scan, and PCWG3 PSA₅₀ response (-58%) at C3D1



Baseline: 6/28/2021

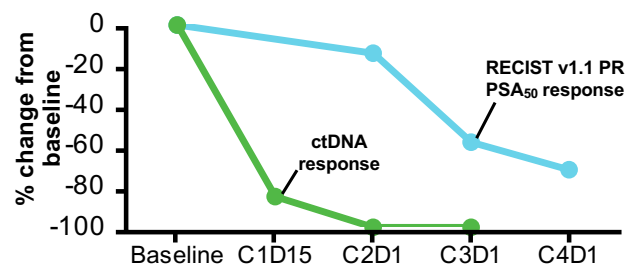


2nd scan: 10/22/2021

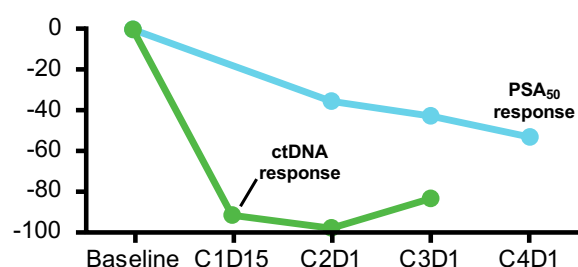


Preliminary evidence of early ctDNA responses and association with clinical response in 5 prostate cancer patients

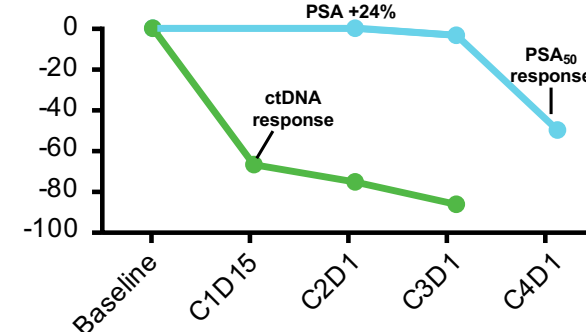
Case #1: 66 y/o; BRCA2m; AZD5305 20 mg
Prior Tx : ADT, abiraterone, and docetaxel



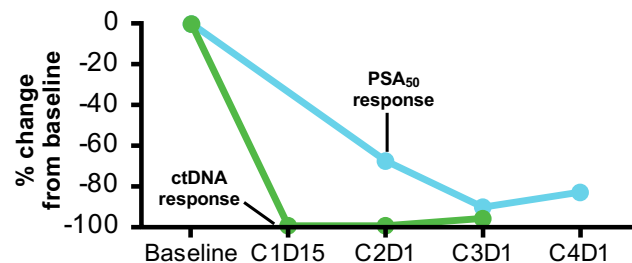
Case #2: 56 y/o; BRCA2m; AZD5305 10 mg
Prior Tx: ADT/abiraterone, and docetaxel



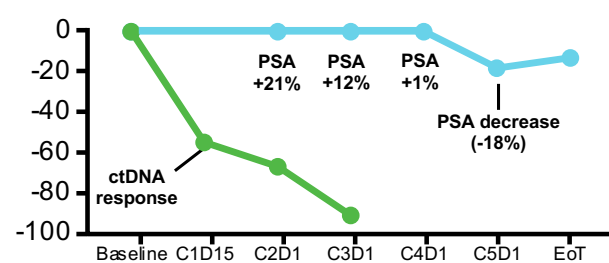
Case #3: 55 y/o; BRCA2m; AZD5305 20 mg
Prior Tx: bicalutamide, ADT, abiraterone, and docetaxel



Case #4: 60 y/o; BRCA2m; AZD5305 20 mg
Prior Tx: docetaxel, enzalutamide, and cabazitaxel



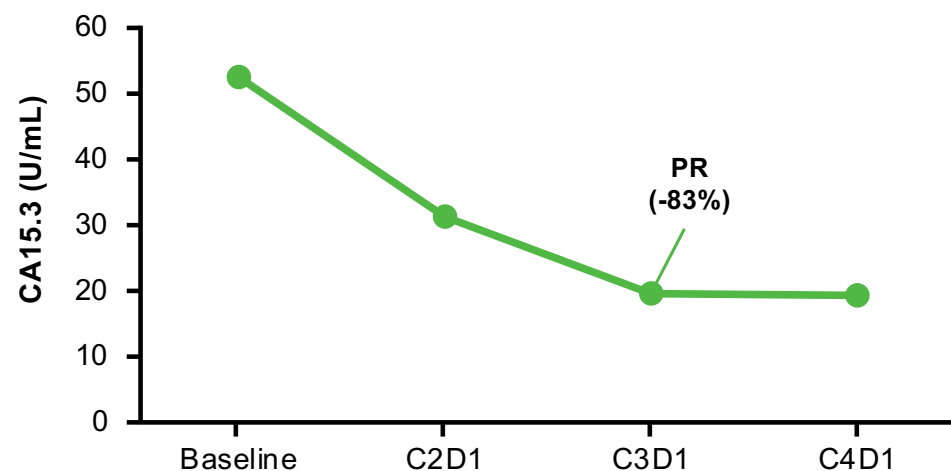
Case #5: 60 y/o; BRCA2m; AZD5305 40 mg
Prior Tx: bicalutamide, docetaxel, and enzalutamide



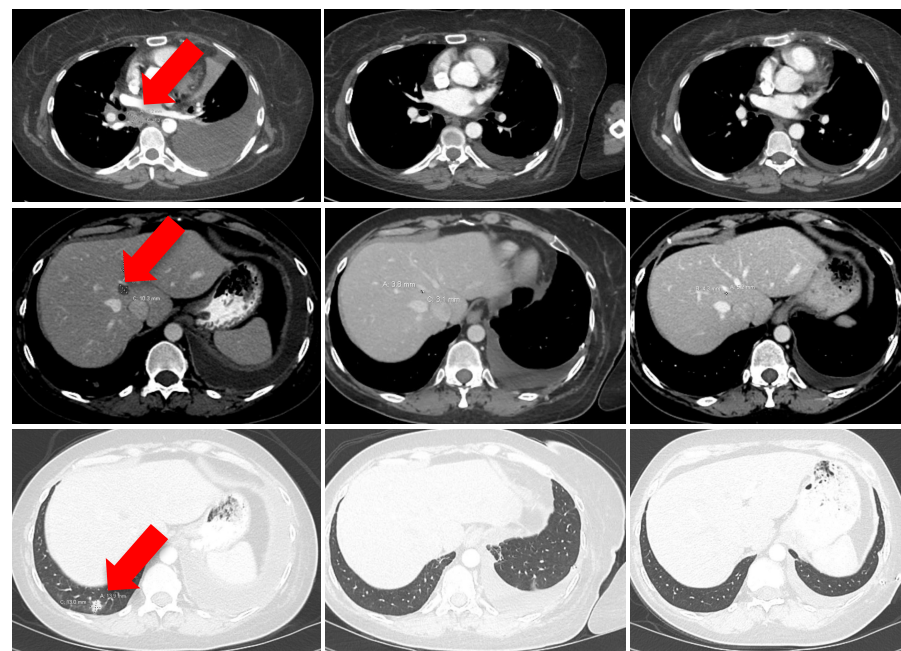
Metastatic breast cancer patient with a RECIST v1.1 confirmed partial response

Case #2: 43 y/o woman with *gRAD51Cm* HR+ breast cancer (AZD5305 40 mg)

- 6 prior lines of therapy*
- RECIST v1.1 PR (-83%), confirmed at second scan (-86.8%)



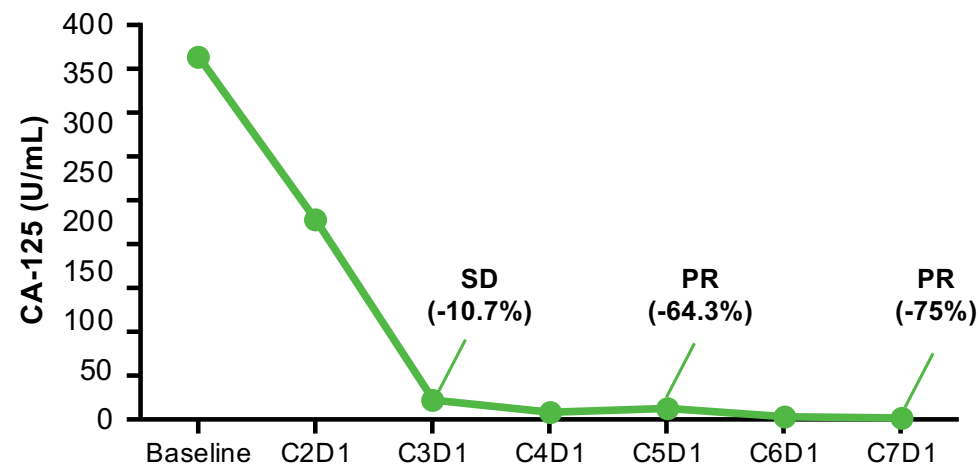
Baseline: 11/11/2021 Scan: 1/18/2022 Scan: 3/15/2022



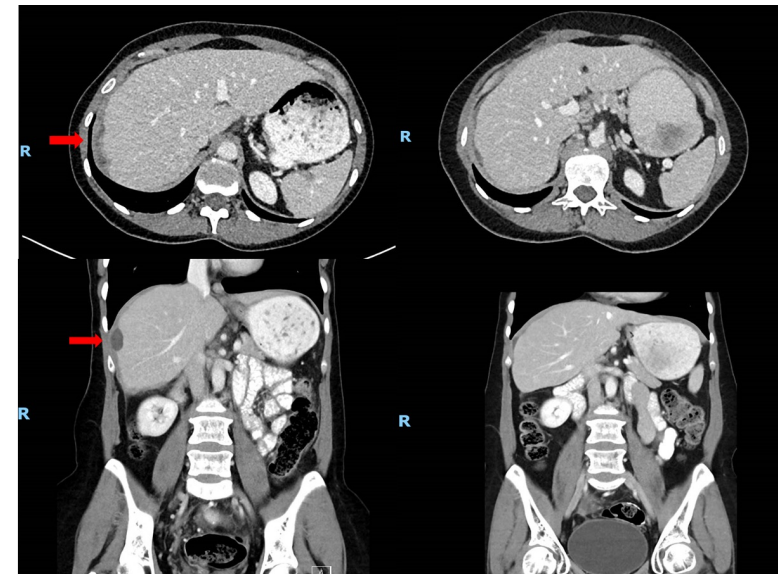
Metastatic ovarian cancer patient with a RECIST v1.1 confirmed partial response

Case #3: 53 y/o woman with *BRCA1*m platinum resistant ovarian cancer (AZD5305 20 mg)

- 3 prior lines of therapy (carbo/pac, LPD/carbo f/b olaparib maintenance*, carbo/gem)
- RECIST v1.1 cPR (-75.0%) at third scan



Baseline: 1/18/2021 Third scan: 7/29/2021



cPR, confirmed Partial Response
*given for 1.6 months, discontinued for nausea

Conclusions and next steps



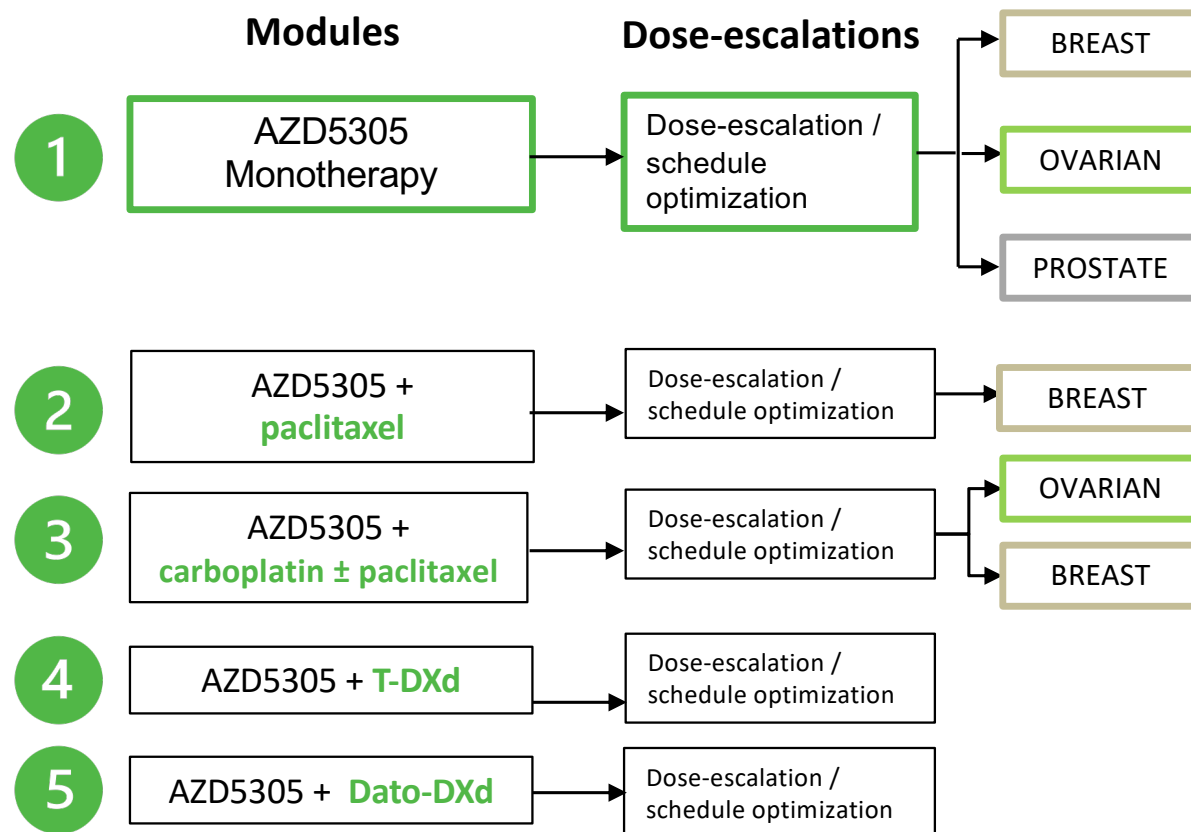
APRIL 8-13 • #AACR22

- AZD5305 is a next generation, highly selective PARP1 inhibitor and trapper
- The excellent pharmacokinetic properties enable significantly improved PK and exposure above target than could be achieved with first generation PARP inhibitors
- Robust and durable target engagement was demonstrated across all dose levels
- Safety profile was highly favorable at all dose levels
- Antitumor activity was observed across selected tumor and molecular subtypes
- Expansion cohorts to evaluate efficacy in PARPi-naïve populations are ongoing
- Dose escalations of rational combinations including trastuzumab deruxtecan* and datopotamab deruxtecan have commenced

PETRA study modular Design

Ongoing and next steps

Indications for expansion cohorts



Acknowledgements



APRIL 8-13 • #AACR22

- **We thank the patients and their carers, as well as the investigators and the site staff who participated in this study.**
- AstraZeneca colleagues: We thank Josh Armenia for the ctDNA analysis; Benedetta Lombardi for the PARylation assay and Nickolas Holoweckyj for the operational aspects of Myriad central testing; Dawid Podgórski, Caroline Kennedy and Yinghui Zhou for the programming and statistical work for the outputs; Chris Wilks, Richard Francis, Suman Nanda, Amy Sant, Ewa Warwick, Julie Stringer, Lauren Cechini, Vanessa Renwick, Muir Russell, Laura Ejikemeuwa, and Joao Simoes for their contributions on the PETRA study; and Mark Albertella, Eli Leo, Anna Staniszevska, Giuditta Illuzzi for their preclinical work
- Medical editing support was provided by Leon Newman, MPharm, PhD, of Ashfield MedComms (London, UK), an Ashfield Health company, and was funded by AstraZeneca. This study was funded by AstraZeneca.
- **Contact information:** tyap@mdanderson.org

Presented at the American Association for Cancer Research (AACR) annual meeting, April 8–13, 2022; New Orleans, LA, USA

Back-up

AACR ANNUAL
American Association MEETING
for Cancer Research 2022 *New Orleans*

APRIL 8-13 • #AACR22

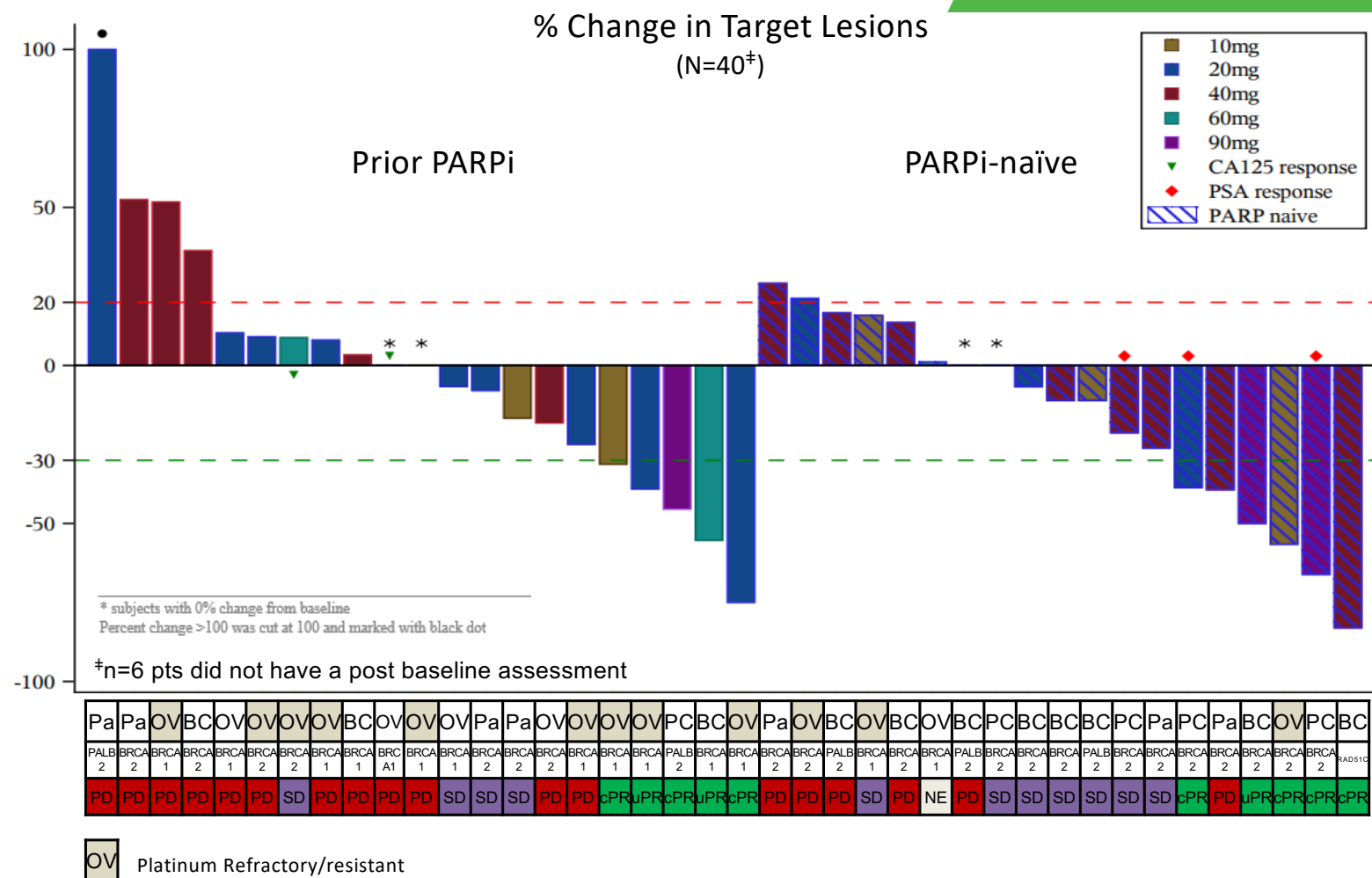
AZD5305-related TEAEs occurring in $\geq 10\%$ of patients overall

TEAE preferred term, (%)	AZD5305 10 mg QD (n=8)		AZD5305 20 mg QD (n=19)		AZD5305 40 mg QD (n=17)		AZD5305 60 mg QD (n=10)		AZD5305 90 mg QD (n=3)		AZD5305 140 mg QD (n=4)		Total (N=61)	
Grade	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All	≥ 3	All
Nausea	0	37.5	0	31.6	0	5.9	0	20	0	33.3	0	0	0	21.3
Neutropenia [†]	12.5	37.5	0	10.5	0	11.8	0	10	66.7	66.7	0	0	4.9	16.4
Thrombocytopenia [†]	12.5	12.5	0	15.8	0	23.5	0	10	0	0	0	0	1.6	14.8
Fatigue and asthenia [†]	0	25	0	10.5	0	17.6	0	20	0	0	0	0	0	14.8
Anemia [†]	25	25	15.8	21.1	5.9	5.9	0	0	33.3	33.3	0	0	11.5	13.1

- AZD5305 was well tolerated across doses
- Rates of nausea and hematological toxicity were low

RECIST v1.1 responses were observed independently of prior PARPi status

	PARPi naïve (n=23)	Prior PARPi (n=23)
PR	5 (21.7%)	5 (21.7%)
confirmed	4	3
unconfirmed	1	2
SD	7 (30.4%)	4 (17.4%)
PD	7 (30.4%)	12 (52.2%)
NE	4 (17.4%)	2 (8.7%)



RECIST v1.1 responses were observed independently of prior PARPi status

APRIL 8-13 • #AACR22

7/9 (78%) patients had a PSA₅₀ response (6 confirmed, 1 unconfirmed)

